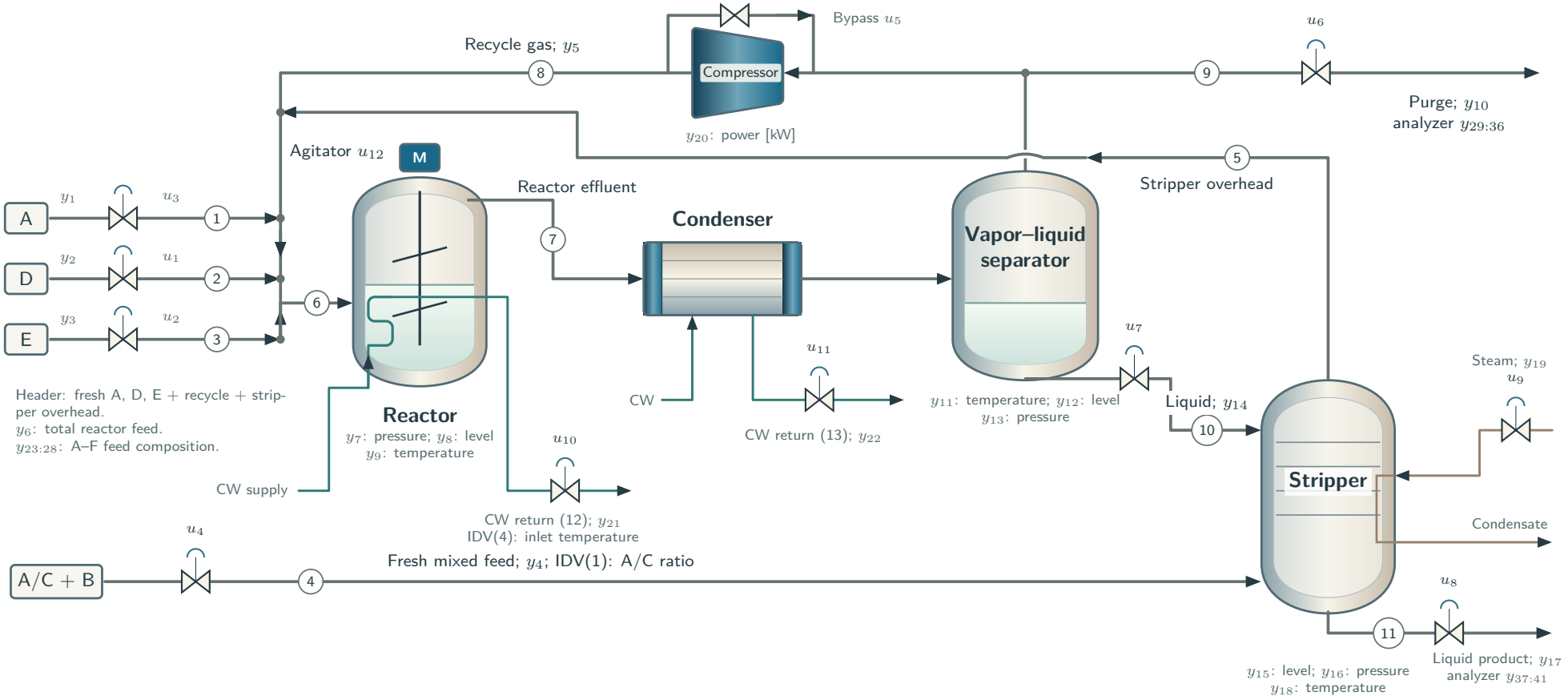
Tennessee Eastman: process, utilities and instruments

Paper section 3.13: *Tennessee Eastman process control problem*



Chemistry and objective
A, C, D, E react; B is inert; G and H are products; F is a byproduct. Reject disturbances while preserving inventories, product rate and quality.

Model boundary
50 states, 12 commands, 41 measurements, six operating modes. The archived native C revision also defines `NX=50`. Downstream G/H separation is outside the benchmark.

Reading the drawing
Gray: process. Teal: CW. Tan: steam/condensate.
Dots: junctions. White bridge: crossing.
Finishes and fluid tints are illustrative.
$u_i = \mathrm{XMV}(i)$; $y_i = \mathrm{XMEAS}(i)$.

Topology checked against Fig. 13 and the Ricker archive; signal indices checked against `temexd_mod.c`. Original explanatory flowsheet, not a construction P&ID. Instrument labels identify measurements, not a prescribed controller.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.